Zhiwei Shao

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Education

Shanghai Jiao Tong University — Shanghai, China
Ph.D. Candidate in Astronomy, GPA: 3.9/4.0 — Sept 2021 – Now

Nanjing University — Nanjing, China
B.S. in Astronomy, Overall GPA: 4.44/5.00, Major GPA: 4.50/5.00 — Sept 2016 – June 2020

- Thesis: "Galaxy Distribution around redMaPPer Clusters in SDSS"
- Advisor: Ying Zu, Shanghai Jiao Tong University

Publications (11 total; 3 first-author)

[1] **Z. Shao**, Y. Zu, A. Salcedo, Y. Lin, Z. Chen, X. Xu, J. Hua, Z. Zhai, J. Aguilar, S. Ahlen, and et al., "Extending the Stellar-to-Halo Mass Relation to Dwarf Galaxies with DESI DR1", *arXiv e-prints*, arXiv:2609.04351, arXiv:2609.04351, Sep. 2026. arXiv: 2609.04351 [astro-ph.GA].

[2] **Z. Shao**, Y. Zu, A. N. Salcedo, J. Wang, X. Yang, D. H. Weinberg, X. Xu, Z. Zhai, Z. Zhang, J. Aguilar, and et al., "Direct Measurement of Galaxy Assembly Bias using DESI DR1 Data", *arXiv e-prints*, arXiv:2510.20896, arXiv:2510.20896, Oct. 2025. arXiv: 2510.20896 [astro-ph.GA].

[3] **Z. Shao**, Y. Zu, and H. Shan, "Is the Large-scale Structure Traced by the BOSS LOWZ Galaxies Consistent with Planck?", *ApJL*, vol. 950, no. 2, L15, p. L15, Jun. 2023. arXiv: 2302.08515 [astro-ph.CO].

[4] L. Clarke, A. E. Shapley, Z. Shao, M. Pascale, R. L. Sanders, N. A. Reddy, T. Jones, H. Katz, N. Lam, S. Karthikeyan, and et al., "The AURORA Survey: Determining the Production Mechanism for OI λ8449 Emission in Star-forming Galaxies at Cosmic Noon", *arXiv e-prints*, arXiv:2609.32045, arXiv:2609.32045, Sep. 2026. arXiv: 2609.32045 [astro-ph.GA].

[5] D. Scholte, F. Cullen, J. Moustakas, H. Zou, A. Saintonge, K. Z. Arellano-Cordova, T. M. Stanton, B. Andrews, J. Sui, J. Aguilar, and et al., "Electron temperature relations and the direct N, O, Ne, S, and Ar abundances of 49 959 star-forming galaxies in DESI data release 2", *MNRAS*, vol. 551, no. 1, stag1381, stag1381, Sep. 2026. arXiv: 2601.02463 [astro-ph.GA].

[6] A. Padawer-Blatt, **Z. Shao**, R. T. Hough, D. Rennehan, R. Barré, V. Saeedzadeh, A. Babul, R. Davé, C. Kobayashi, W. Cui, and et al., "Core to Cosmic Edge: SIMBA-C's New Take on Abundance Profiles in the Intragroup Medium at z = 0", *Universe*, vol. 11, no. 2, 47, p. 47, Feb. 2025. arXiv: 2502.04657 [astro-ph.GA].

[7] R. T. Hough, **Z. Shao**, W. Cui, S. I. Loubser, A. Babul, R. Davé, D. Rennehan, and C. Kobayashi, "Simba-C: the evolution of the thermal and chemical properties in the intragroup medium", *MNRAS*, vol. 532, no. 1, pp. 476–495, Jul. 2024. arXiv: 2406.04761 [astro-ph.GA].

[8] V. Saeedzadeh, S. L. Jung, D. Rennehan, A. Babul, M. Tremmel, T. R. Quinn, **Z. Shao**, P. Sharma, L. Mayer, E. O'Sullivan, and S. I. Loubser, "Cool and gusty, with a chance of rain: Dynamics of multiphase CGM around massive galaxies in the ROMULUS simulations", *MNRAS*, Sep. 2023. arXiv: 2304.03798 [astro-ph.GA].

[9] X. Chen, Y. Zu, **Z. Shao**, and H. Shan, "The sphere of influence of the bright central galaxies in the diffuse light of SDSS clusters", *MNRAS*, vol. 514, no. 2, pp. 2692–2706, Aug. 2022. arXiv: 2112.03934 [astro-ph.GA].

[10] Y. Zu, Y. Song, **Z. Shao**, X. Chen, Y. Zheng, H. Gao, Y. Yu, H. Shan, and Y. Jing, "Strong conformity and assembly bias: towards a physical understanding of the galaxy-halo connection in SDSS clusters", *MNRAS*, vol. 511, no. 2, pp. 1789–1807, Apr. 2022. arXiv: 2108.06790 [astro-ph.CO].

[11] Y. Zu, H. Shan, J. Zhang, S. Singh, **Z. Shao**, X. Chen, J. Yao, J. B. Golden-Marx, W. Cui, E. Jullo, J.-P. Kneib, P. Zhang, and X. Yang, "Does concentration drive the scatter in the stellar-to-halo mass relation of galaxy clusters?", *MNRAS*, vol. 505, no. 4, pp. 5117–5128, Aug. 2021. arXiv: 2012.08629 [astro-ph.CO].

Talks

- DESI Winter Collaboration Meeting, Tucson, USA — Dec 2025
- Gravitational Lensing Workshop, Shanxi, China — Sept 2024
- "Small Galaxies, Cosmic Questions - II", Durham, UK — July 2024
- The 32nd Texas Symposium on Relativistic Astrophysics, Shanghai, China — Dec 2023
- The 2nd Shanghai Assembly on Cosmology and Structure Formation, Shanghai, China — Nov 2023
- DESI July 2023 Hybrid Collaboration Meeting, Durham, UK (spotlight talk) — July 2023
- 10th Workshop on Gravitational Lensing, Qingdao, China — June 2023
- The 1st KIAA Forum on Gas in Galaxies for Early Career Scientists, KIAA, Peking University, China (online) — Nov 2021

Honors and Awards

- T.D. Lee Scholarship — 2024
- Minghong Scholarship (First Class) — 2023
- Yangyang Scholarship — 2023
- Outstanding Graduate of Nanjing University — 2020
- Member of NJU Elite Project — 2016 – 2020
- Elite Project Scholarship — 2017, 2019
- People's Scholarship — 2017 – 2019
- Annual Scholarship of NAOC, CAS — 2018
- Excellent Student — 2017

Outreach

- **Volunteering** at Guangqi Observatory, Shanghai Jiao Tong University — 2021-2024
  Helping to set up telescopes, guiding visitors, and sharing basic astronomy knowledge every month.
- **Teaching** at Qinhuai 2nd Experimental Primary School — Spring 2017
  Teaching pupils elementary astronomy knowledge.